

Marking scheme of Calculus for Engineerers I

Question 1.

- a. Verify whether the following numerical series converges or not (2Marks)

$$\sum_{k=1}^{+\infty} \left(\frac{2k-5}{4k+1} \right)^k$$

Answer:

$$\lim_{k \rightarrow \infty} \sqrt[k]{\left(\frac{2k-5}{4k+1} \right)^k} = \lim_{k \rightarrow \infty} \left(\frac{2k-5}{4k+1} \right) = \frac{2}{4} = \frac{1}{2} \quad /1\text{Mark}$$

As this limit is less than one, then the above series converged /1Mark

- b. Find the real value of x such that the inverse of the matrix $A = \begin{bmatrix} x & 1 \\ 2 & 5 \end{bmatrix}$ does exist. (2Marks)

Answer: From the definition of the existence of inverse of the matrix we have:

$$|A| = \det(A) = \begin{vmatrix} x & 1 \\ 2 & 5 \end{vmatrix} \neq 0 \quad /1 \text{ mark}$$

$$\Leftrightarrow 5x - 2 \neq 0 \quad /0.5 \text{ mark}$$

$$\Leftrightarrow x \neq \frac{2}{5} \quad \text{or } x \in \mathbb{R} - \left\{ \frac{2}{5} \right\}. \quad /0.5 \text{ mark}$$

- c. Compute $\frac{dy}{dx}$ for $t = 1$, if (2Marks)

$$\begin{cases} x = t \ln t \\ y = \frac{\ln t}{t} \end{cases}$$

Answer: By definition we get: $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \quad /0.5\text{mark}$

$$\frac{dy}{dt} = \frac{d\left(\frac{\ln t}{t}\right)}{dt} = \frac{1 - \ln t}{t^2} \quad \text{and} \quad \frac{dx}{dt} = \frac{d(t \ln t)}{dt} = 1 + \ln t \quad /0.5\text{mark}$$

$$\text{Then } \frac{dy}{dx} = \frac{\frac{1 - \ln t}{t^2}}{1 + \ln t} = \frac{1 - \ln t}{t^2} \times \frac{1}{1 + \ln t} = \frac{1 - \ln t}{t^2(1 + \ln t)} \quad /0.5\text{mark}$$

$$\left. \frac{dy}{dx} \right|_{t=1} = \frac{1}{1^2} = 1.$$

/0.5mark

d. Find y in terms of x given that:

(3marks)

$$5 \log_2 y - 3 \log_2 (x + 4) = 2 \log_2 y + 3 \log_2 x \quad \forall x, y \in \mathbb{R} - \{0\}.$$

Answer:

$$\Leftrightarrow 5 \log_2 y - 2 \log_2 y = 3 \log_2 (x + 4) + 3 \log_2 x$$

$$\Leftrightarrow 3 \log_2 y = 3 \log_2 (x + 4) + 3 \log_2 x \quad /0.5mark$$

$$\Leftrightarrow 3 \log_2 y = 3 \log_2 x(x + 4) \quad /0.5mark$$

$$\Leftrightarrow \log_2 y = \log_2 x(x + 4) \quad /1mark$$

$$\Leftrightarrow y = x^2 + 4x \quad /0.5mark$$

$$\frac{dy}{dx} = 2x + 4 \quad /0.5mark$$

e. Given $f(x) = \frac{x+1}{x-1}$

i) Calculate the limits around the boundaries of domain of definitions and deduce the equations of all possible asymptotes (3 Marks)

ii) Find its integral on this interval $[2, 3]$ (2 Marks)

Answer:

i) $dom f = \mathbb{R} - \{1\}$ /0.5mark

$$\lim_{x \rightarrow 1^-} \left(\frac{x+1}{x-1} \right) = -\infty \text{ and } \lim_{x \rightarrow 1^+} \left(\frac{x+1}{x-1} \right) = \infty \quad /0.5mark$$

$\Rightarrow$ that the function has vertical Asymptote i.e $V.A \equiv x = 1$ /0.5mark

$$\lim_{x \rightarrow \infty} \frac{x+1}{x-1} = 1 \quad /0.5mark$$

$\Rightarrow$ that the function has Horizontal Asymptote i.e the $H.A = y = 1$ /0.5mark

No oblique Asymptotes

/0.5mark

ii)

$$\int_2^3 f(x) = \int_2^3 \frac{x+1}{x-1} dx = \int_2^3 \frac{x-1+2}{x-1} dx = \int_2^3 \frac{x-1}{x-1} dx + \int_2^3 \frac{2}{x-1} dx \quad /0.5mark$$

$$= \int_2^3 dx + \int_2^3 \frac{2}{x-1} dx \quad /0.5mark$$

$$= [x]_2^3 + 2[\ln(x-1)]_2^3 \quad /0.5mark$$

$$= (3-2) + 2(\ln 2 - \ln 1) = 1 + 2 \ln 2 = 1 + \ln 4 \quad /0.5mark$$

f. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined as $T(\vec{x}) = \langle x_2, x_1, x_1 + x_2 \rangle$. Show that T is a linear operator? (3Marks)

Answer:

T is a linear operator iff $L(\alpha\vec{x} + \beta\vec{y}) = \alpha L(\vec{x}) + \beta L(\vec{y}), \forall \vec{x}, \vec{y} \in \mathbb{R}^2$, and $\forall \alpha, \beta$ are scalars /0.5mark

Verification:

$$T(\alpha\vec{x} + \beta\vec{y}) = T\left(\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \beta \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} \alpha x_1 \\ \alpha x_2 \end{bmatrix} + \begin{bmatrix} \beta y_1 \\ \beta y_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} \alpha x_1 + \beta y_1 \\ \alpha x_2 + \beta y_2 \end{bmatrix}\right)$$

/0.5mark

$$= \begin{bmatrix} \alpha x_2 + \beta y_2 \\ \alpha x_1 + \beta y_1 \\ (\alpha x_1 + \beta y_1) + (\alpha x_2 + \beta y_2) \end{bmatrix} = \begin{bmatrix} \alpha x_2 + \beta y_2 \\ \alpha x_1 + \beta y_1 \\ \alpha(x_1 + x_2) + \beta(y_1 + y_2) \end{bmatrix} \quad /0.5mark$$

$$= \begin{bmatrix} \alpha x_2 \\ \alpha x_1 \\ \alpha(x_1 + x_2) \end{bmatrix} + \begin{bmatrix} \beta y_2 \\ \beta y_1 \\ \beta(y_1 + y_2) \end{bmatrix} \quad /0.5mark$$

$$= \alpha \begin{bmatrix} x_2 \\ x_1 \\ (x_1 + x_2) \end{bmatrix} + \beta \begin{bmatrix} y_2 \\ y_1 \\ (y_1 + y_2) \end{bmatrix} = \alpha T(\vec{x}) + \beta T(\vec{y}) \quad /1mark$$

Then, we confirm that T is a linear operator

g. Show that the $f(x) = \sin x$ satisfies the Rolle's Theorem everywhere? (3Marks)

Answer:

$f(x) = \sin x$ is both continuous and differentiable everywhere, so that $f(x) = \sin x$ is continuous on $[0, 2\pi]$ and differentiable on $(0, 2\pi)$ /1mark

$f(0) = \sin 0 = 0$ and $f(2\pi) = \sin 2\pi = 0$ /0.5mark

So that f satisfies the hypotheses of Rolle's Theorem on the interval $[0, 2\pi]$.

Thus, we are guaranteed that there is at least one point

$c \in (a, b)$ such that $f'(c) = 0$. /0.5mark

$f'(c) = 0 \Leftrightarrow \cos c = 0 \rightarrow c = \frac{\pi}{2}$ or $c = \frac{3\pi}{2}$ /1mark

SECTION B / 30 Marks

QUESTION 2.

Consider the function defined by $y = x^2 + 1$ and the line $D \equiv x + y = 3$ on the interval $[0, 2]$.

- Find the intersection points between the line and the curve of the function. (5marks)
- Draw the function and the line in the same plane (5marks)
- Use integration to calculate the area bounded by the two curves (5marks)

Answer:

- The intersection points of D and P are given by the solution of the

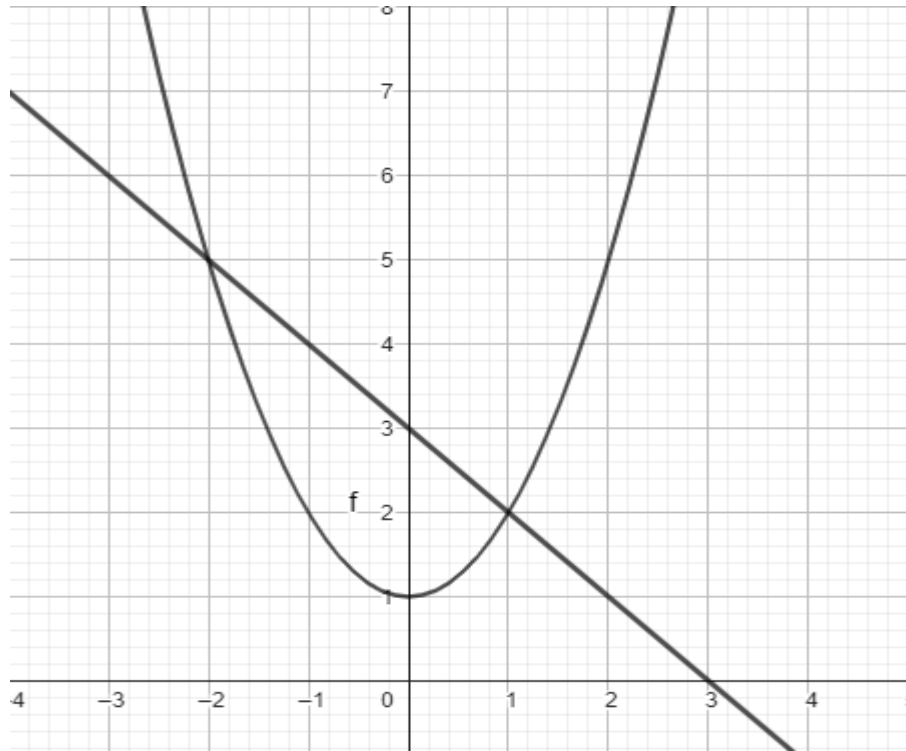
$$\text{system: } \begin{cases} y = x^2 + 1 \\ x + y = 3 \Rightarrow y = 3 - x \end{cases} \quad /1mark$$

The equation $y = x^2 + 1$ becomes: $3 - x = x^2 + 1 \Rightarrow x^2 + x - 2 = 0$ /1mark

The solution of the quadratic equation are: $x = -2$ and $x = 1$ /1mark

Then the intersection points are $(-2,5)$ and $(1,2)$. /2marks

b.



For the straight line. /1mark

For the parabola /1mark

For well scaled axes /1mark

For x and y-intercepts /1mark

For the intersection of the parabola and the straight line. /1mark

c. Let $f(x) = x^2 + 1$ and $g(x) = 3 - x$;

$$\text{then } S = \int_0^1 (g(x) - f(x)) dx + \left| \int_1^2 (g(x) - f(x)) dx \right| \quad /2marks$$

Or $g(x) - f(x) = -x^2 - x + 2.$

/1mark

$$S = \left[-\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_0^1 + \left| \left[-\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_1^2 \right|$$

/1mark

$$= (-1/3 - 1/2 + 2) + \left| \left(-\frac{8}{3} - 2 + 4 \right) - \left(-1/3 - 1/2 + 2 \right) \right| = -5/6 + 2 + 11/6 = 3 \text{ au.}$$

/1mark

QUESTION 3.

- a. Find the interval of convergence and radius of convergence of the power series
(5Marks)

$$\sum_{k=0}^{+\infty} \frac{(-1)^k (x-1)^k}{3^{k(k+1)}}$$

- b. Find the eigenvalues and the corresponding eigenvectors for the following matrix

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

(10marks)

Answer:

- a. We apply the ratio test for absolute convergence.

$$q = \lim_{k \rightarrow \infty} \frac{|u_{k+1}|}{|u_k|} = \lim_{k \rightarrow \infty} \left| \frac{(x-1)^{k+1}}{3^{k+1}(k+2)} \cdot \frac{3^k(k+1)}{(x-1)^k} \right| = \lim_{k \rightarrow \infty} \left| x-1 \right| \left(\frac{k+1}{k+2} \right) = |x-1|$$

/1mark

The series converges absolutely if $q < 1 \Leftrightarrow |x-1| < 1 \Leftrightarrow -1 < x-1 < 1$ or $0 < x < 2$, and diverges if $x > 2$ and $x < 0$

/1mark

The ratio test fails when $x = 0$, and $x = 2$ need separate analyse.

Substituting $x = 0$ in a given series yield:

$$\sum_{k=0}^{+\infty} \frac{1^k}{3^k(k+1)} = \sum_{k=0}^{+\infty} \frac{1}{3^k(k+1)}$$

Which is the convergence series

/0.5mark

Substituting $x = 2$ in a given series yield:

$$\sum_{k=0}^{+\infty} \frac{(-1)^k}{3^k(k+1)}$$

Which is absolutely convergent series at $x = 2$. **/0.5mark**

Thus the interval of convergence for the given series is $[0, 2]$ and the radius of convergence is $R = 1$. **/2marks**

b. The characteristic equation:

$$\det(A - \lambda I) = 0 \Leftrightarrow \begin{vmatrix} 1-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix} = 0 \quad \textbf{/2marks}$$

$$\rightarrow \lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 3 \quad \textbf{/2marks}$$

For $\lambda = 0$, we have $(A - 0I)x = 0 \Leftrightarrow \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow x = y, y = z$

$$\vec{x} = x \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \forall x \in \mathbb{R}^* \quad \textbf{/1mark}$$

Therefore $\lambda = 0$ is an Eigen-value of a matrix A and $\vec{x} = x \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \forall x \in \mathbb{R}^*$ is an

Eigen-vector or characteristic. **/1mark**

For $\lambda = 1$, we have $(A - 1I)x = 0 \Leftrightarrow \begin{bmatrix} 0 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow x = -z, y = 0$

$$\vec{x} = x \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \forall x \in \mathbb{R}^* \quad \textbf{/1mark}$$

Therefore $\lambda = 1$ is an Eigen-value of a matrix A and $\vec{x} = x \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\forall x \in \mathbb{R}^*$ is an

Eigen-vector or characteristic. /1mark

For $\lambda = 3$, we have $(A - 1I)x = 0 \leftrightarrow \begin{bmatrix} -2 & -1 & 0 \\ -1 & -1 & -1 \\ 0 & -1 & -2 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \rightarrow x = z, y = -2x$

$\vec{x} = x \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$, $\forall x \in \mathbb{R}^*$ **1mark**

Therefore $\lambda = 1$ is an Eigen-value of a matrix A and $\vec{x} = x \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$, $\forall x \in \mathbb{R}^*$ is an

Eigen-vector or characteristic. /1mark

QUESTION 4.

a. Consider the matrices $J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

Show that $A^n = 3^n I_3 + a_n J$ where a_n is a real number that you determine.

(7marks)

b. A box with an open top is to be constructed from a rectangular piece of cardboard with dimensions 14 meter by 22 meter by cutting out equal squares of side x at each corner and then folding up the sides. Express the volume V of the box as a function of x and find the value of x for which the volume is maximal. **(8marks)**

Answer:

$$\text{a. } A^2 = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} \cdot \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 9 & 6 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} +$$

$$\begin{pmatrix} 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 3^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + 2 \times 3 \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 3^2 I_3 + 2 \times 3 J$$

/1mark

$$A^3 = \begin{pmatrix} 9 & 6 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} \cdot \begin{pmatrix} 9 & 6 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} = \begin{pmatrix} 27 & 27 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{pmatrix} = \begin{pmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{pmatrix} +$$

$$\begin{pmatrix} 0 & 27 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 3^3 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + 3 \times 3^2 \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 3^3 I_3 + 3 \times 3^2 J$$

/1mark

$$A^4 = \begin{pmatrix} 27 & 27 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{pmatrix} \cdot \begin{pmatrix} 27 & 27 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{pmatrix} = \begin{pmatrix} 81 & 108 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{pmatrix} =$$

$$\begin{pmatrix} 81 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{pmatrix} + \begin{pmatrix} 0 & 108 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 3^4 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + 4 \times 3^3 \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} =$$

$$3^4 I_3 + 4 \times 3^3 J \quad \text{color: red; /1mark}$$

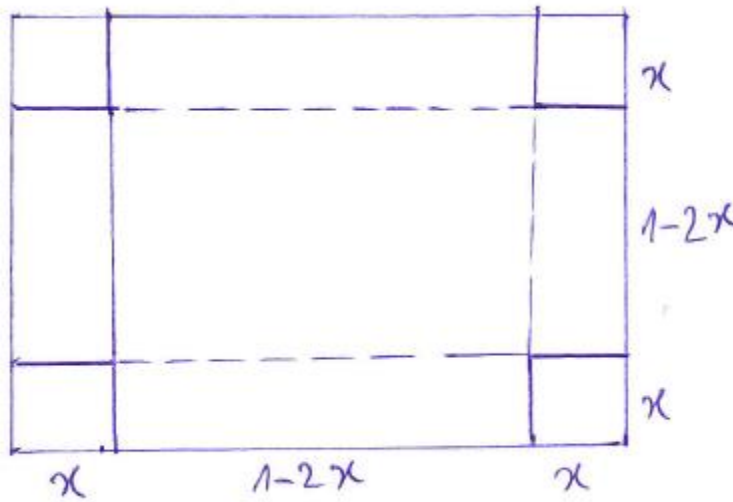
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$$A^n = \begin{pmatrix} 3^n & n \cdot 3^{n-1} & 0 \\ 0 & 3^n & 0 \\ 0 & 0 & 3^n \end{pmatrix} = \begin{pmatrix} 3^n & n \cdot 3^{n-1} & 0 \\ 0 & 3^n & 0 \\ 0 & 0 & 3^n \end{pmatrix} + n \cdot 3^{n-1} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

/2marks

$$a_n = n \cdot 3^{n-1} \quad \text{color: red; /2marks}$$

b.



/1mark

$$V = (1 - 2x)^2 x = 4x^3 - 4x^2 + x \quad /2\text{marks}$$

The maximum is reached when $\frac{dV}{dx} = 0$ and $\frac{d^2V}{dx^2} < 0$ /1mark

$$\frac{dV}{dx} = 0 \Leftrightarrow 12x^2 - 8x + 1 = 0 \Rightarrow x = \frac{1}{2} \text{ or } x = \frac{1}{6} \quad /2\text{marks}$$

$$\frac{d^2V}{dx^2} = 24x - 8$$

$$\left. \frac{d^2V}{dx^2} \right|_{x=\frac{1}{2}} = 24 \left(\frac{1}{2} \right) - 8 = 4 > 0 \Rightarrow$$

for this point we do not reach to maximum /1mark

$$\left. \frac{d^2V}{dx^2} \right|_{x=\frac{1}{6}} = 24 \left(\frac{1}{6} \right) - 8 = -4 < 0 \Rightarrow$$

for this point we reach to maximum /1mark